- **FLOWERS** There is some simply aesthetic application and an example of this is the use of gene recognition and transfer techniques to improve the color, smell, size and other features of flowers. Similarly, biotech has been used to make improvement to other common ornamental plants, in particular, shrubs and trees. Some of these changes are similar to those made to crops, such as enhancing cold confrontation of a breed of tropical plant, so it can be grown in northern gardens.

- **BIOFUELS :** The agricultural industry plays a big role in the biofuels industry, as long as the feedstock's for fermentation and cleansing of bio-oil, bio-diesel and bio-ethanol. Genetic engineering and enzyme optimization technique are being used to develop improved quality feedstocks for more efficient change and higher BTU outputs of the resulting fuel products.

MEDICAL BIOTECHNOLOGY

- The study of the human body, its diseases, and the prevention of those diseases.
- Focuses on the use of living cells, microorganisms, and cell materials to research human diseases and develop pharmaceutical drugs to treat and/or prevent medical maladies.
- Most often applied to the pharmaceutical industry (where biotechnology corporations have grown in both size and importance.)



MEDICAL BIOTECHNOLOGY (CONT)

- Medical Biotechnology also covers pharmacogenomic and genetic testing.
- Pharmacogenomics - - the branch of biotechnology that tests the response that the human body has to those drugs that are created.
- Genetic testing - - the biotechnological research of inherited diseases based on an individual's genealogy.



AGRICULTURAL BIOTECHNOLOGY

- Focused on improving and changing plants, animals, and microorganisms.
- Scientists use current understanding of DNA to modify genetic makeup of organisms.
- Most often used in the genetic engineering of crops.
- Genes are transferred between crops to enhance a variety of qualities: taste, nutrition, appearance, longevity, and ability to withstand climate changes & natural predators.



INDUSTRIAL BIOTECHNOLOGY

- Uses plants and marine organisms to create industrial goods.
- Microorganisms and components of cells used to improve the production of a variety of goods (Ex: chemicals, textiles, biofuels, and paper)
- Research conducted to improve the quality & environmental impact of the goods
- Industrial technology is considered the largest branch of biotechnology.



Industrial Biotechnology:

- It, often referred to as "white biotechnology", is a collection of scientific techniques and technologies used to improve both the efficiency and environmental footprint of modern industrial production.
- Microbial technology constitutes the core of Industrial Biotechnology.
- Microbial technology refers to the use of microbes to obtain a product or service of economic value. It is also called as fermentation.

Activities In Industrial Microbiology:

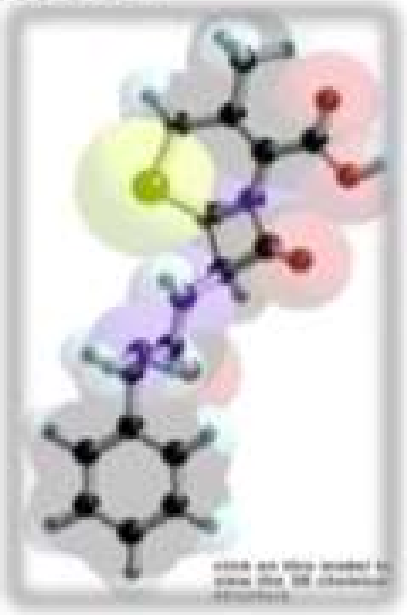
- Isolation of microorganisms from nature,
- Their screening for product formation,
- Improvement of product yields,
- Maintenance of cultures,
- Mass culture using bioreactors, and
- Recovery of products or services.

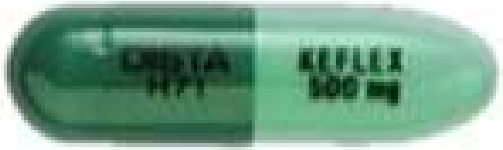
Applications:

- Metabolite production.
- Anaerobic digestion (for methane production).
- Waste treatment (both organic and industrial).
- Production of biocontrol agents, and
- Fermentation of food products.
- Bio based fuel &energy.

Metabolite Production:

- Microorganisms produce a number of metabolites during their growth using cheap substrates.
- Acetone-butanol,
- Alcohol,
- Antibiotics,
- Enzymes,
- Organic acids
- Vitamins.





Biomass production:

- Microorganisms employed to enhance the availability of nutrients, viz., nitrogen (by fixing atmospheric N^2) and phosphorus (by solubilizing soil phosphorus), to the crops are called biofertilizers.
- *Rhizobium spp.*
- Blue-green Algae and Azolla.
- Azotobacter and Azospirillum.

RECOVERY OF METALS :

- Microbes can be employed to recover valuable metals from low grade ores and also from dilute solutions for which the conventional metallurgical processes are uneconomical and, generally, rather polluting.

E.g

- *leaching of copper as copper sulphate from ores.*
- Desulphurization of Coals by *Thiobacillus spp.*

Production of biocontrol agents:

- Use of microorganisms to control **insect pests, pathogens or weeds constitutes biological control, and the biological agents employed for this purpose are called biocontrol agents.**
- **Bioinsecticides-**
- **Bioherbicides-**



Fermentation of food products:



Bio based fuel & energy:

- Ethanol currently produced by fermenting grain (old technology).
- Cellulose enzyme technology allows conversion of crop residues (stems, leaves and hulls) to ethanol.
- Results in reduced CO_2 emissions by more than 90% (compared to oil).
- Allows for greater domestic energy production and it uses a renewable feedstock.



Benefits:

- Low input of substrate.
- High rate of output.
- Friendly to environment.
- Renewable.
- Increased efficiency.